Pipeline Validation: POS-Tagger Migration and Associated Fixes

TODO: author name

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Abstract

This note documents a validation pass performed on the Macedonian-language stylometric pipeline after its part-of-speech tagger was migrated from spaCy (`mk_core_news_lg`) to CLASSLA-Stanza. It covers (i) how much the downstream Louvain community structure on the semantic knowledge graph (SKG) changed as a result of the tagger switch, separated from ordinary seed-to-seed algorithmic noise, including the diagnosis and subsequent fix of a community-detection artifact (author-similarity edges alone defining a 13-author community with almost no word support) and the seed-stability tradeoff that fix introduces; (ii) two correctness bugs found and fixed during the pass, one in rhyme-ending extraction and one in the style-transfer candidate-selection fallback tier; (iii) two metric additions — a length-normalized lexical-diversity measure and stanza-level structural features — added to the per-author style profiles; and (iv) fixes to the prompt construction of the LLM style-transfer pipeline and its switch to Gemma 4. The distillation and reinforcement-learning stages built on that pipeline are reported separately (`rl_report.tex`).

1 Background: the tagger migration

The pipeline originally tagged the corpus (2,711 Macedonian poems and song lyrics, 24 authors after the ≥ 5-poem eligibility filter) with spaCy's `mk_core_news_lg` model. This was replaced with CLASSLA-Stanza, which lemmatizes inflection correctly and additionally exposes MULTEXT-East `xpos`/`feats` morphology, enabling the feats-keyed morphological lookup used by style transfer. The full pipeline — POS tagging, TF-IDF profiling, SKG construction and enrichment, embeddings, and graph analytics — was rerun end to end against the new tags. This note asks: how much did that rebuild actually change the pipeline's outputs, and were any of those changes symptoms of pre-existing bugs rather than the tagger switch itself?

2 Louvain community stability under the tagger switch

2.1 Setup

Author-similarity communities are computed by `graph_analytics.py` via `community.best_partition(G, weight='weight',random_state=42)` on the SKG (`skg_final.gexf`). Both the pre-switch (spaCy) and post-switch (CLASSLA) versions of `graph_analytics.py` call Louvain with these exact same parameters, so a direct diff of the two partitions is a fair comparison, not confounded by a methodology change alongside the tagger change. The pre-switch graph and analytics output were recoverable from git history (`git` commit `5f03983`, carried forward unchanged by the pure file-reorganization commit `363009a`); the post-switch version is the current repository state (commit `41bb49e` onward).

2.2 Direct diff against the pre-switch partition

Both partitions cover the same 24 authors and both settle on 6 communities. Comparing author community labels directly (after optimal label alignment) gives:

Metric	Value
Adjusted Rand Index (ARI)	0.221
Normalized Mutual Information (NMI)	0.483

An ARI of 1 would mean the partitions are identical (up to relabeling); an ARI near 0 means the agreement is no better than chance. 0.221 indicates a substantial reshuffling — but Louvain is itself stochastic in the order it visits nodes, so this number alone cannot distinguish "the tagger switch changed the graph structure" from "Louvain would have landed somewhere this different anyway, even on an unchanged graph, under a different run."

2.3 Establishing a noise floor

To separate the two, Louvain was re-run 30 times on the *current* (post-switch) graph with `random_state` $\in \{0, \ldots, 29\}$, holding the graph itself fixed. Each run was compared to the documented `random_state=42` reference partition:

Comparison	Min	Mean	Median
Each of 30 seeds vs. the seed $= 42$ reference (ARI)	0.515	0.877	0.906
All pairs among the 30 seeds (ARI, 60 sampled pairs)	0.469	0.819	—

The number of communities Louvain settles on also varies by seed alone (4, 5, or 6 communities across the 30 runs).

2.4 Conclusion: a real change, not just seed noise

Seed-only variability on a fixed graph floors at ARI ≈ 0.47–0.5 in the worst observed case. The pre/post-switch ARI (0.221) sits *below* that floor. The tagger switch therefore produced a partition shift larger than ordinary Louvain run-to-run noise would account for — a genuine, if partial, structural change, not an artifact of re-running the algorithm once.

The change is concentrated among the "word-driven" authors (see §2.5 for the exception). Three of the six pre-switch communities (covering 14 of the 24 authors) dissolved and redistributed across different post-switch communities — e.g. the pre-switch cluster {Рацин, Поповски, Коцев, Линдро, Лазаров} split into four different post-switch communities. This is the expected effect of a tagger change: different lemmatization directly perturbs the TF-IDF and co-occurrence edges those communities are built from.

2.5 Diagnosis: Community 0, a stable but low-information clique

The current partition's largest community ("Community 0") groups 13 of the 24 authors around only 2 surviving word nodes:

> Петре М. Андреевски, Влада Урошевиќ, Блаже Конески, Славко Јаневски, Ацо Шопов, Матеја Матевски, Богомил Ѓузел, Адем Гајтани, Веле Смилевски, Пијан Славеј, Ефтим Клетников, Ристо Лазаров, Ива Петреска

Seed stability. All $\binom{13}{2} = 78$ pairs of these authors co-clustered together in *every one* of the 30 seed runs described above — the single most reproducible feature of the entire partition, far above the general seed-noise floor.

Why it forms. Breaking down each author's SKG weighted degree by edge type shows why: every Community-0 author draws 40–70% of its weighted degree from direct `author_similarity` edges (author ↔ author Jaccard-vocabulary-overlap edges), versus 86–100% from word-mediated (`distinctive`/`cooccurrence`) edges for every author outside the community. Community 0 is not a set of authors Louvain found to write alike through shared vocabulary; it is the set of authors whose word nodes did not survive strongly into the graph (consistent with the SKG's frequency floors and node cap), leaving the direct author-similarity edges as the dominant signal available to place them by.

Not new to the tagger switch. The same phenomenon already existed pre-switch, with a 9-author core — Андреевски, Урошевиќ, Конески, Јаневски, Ѓузел, Гајтани, Славеј, Клетников, Петреска — all a subset of today's 13, each already showing 13–47% author-similarity weight share before the tagger changed. The switch grew this core from 9 to 13 authors (adding Шопов, Матевски, Смилевски, Лазаров), it did not create it.

Interpretation. Community 0 should be read as "authors the graph has too little word-level signal to place elsewhere," not as evidence of shared literary style — and, as described next, was subsequently removed by changing what Louvain is allowed to see.

2.6 Fix: excluding `author_similarity` edges from the partitioning graph

`graph_analytics.py` now runs Louvain on a copy of the SKG with every `author_similarity` edge removed, keeping `cooccurrence`, `semantic_similarity`, and `distinctive` edges — i.e. only edges mediated by an actual word — as the partitioning signal. `author_similarity` edges are still computed and still present in `skg_final.gexf` for every other purpose (PageRank, weighted degree, visualization); they are excluded only from the graph Louvain sees, since §2.5 showed they were sufficient on their own to define a community with no word-level support behind it. The direct author-similarity signal remains available where it was always meant to be used as an auxiliary/fallback source — in `candidate_selection.author_similarity()`, which already computed it independently of the graph, for `tier_cross_author_similar`.

Result. Community 0 no longer exists. Louvain on the restricted graph settles on 5 communities instead of 6, and every one now has substantial word support (49–102 word nodes each, versus the old Community 0's 2). The 13 authors previously grouped by author-similarity edges alone are now distributed across all 5 communities on genuine vocabulary evidence:

Former Community-0 author	New community (by word evidence)
Блаже Конески, Славко Јаневски, Ацо Шопов	Community 2 (with Кочо Рацин)
Влада Урошевиќ, Пијан Славеј, Ефтим Клетников	Community 1
Петре М. Андреевски, Богомил Ѓузел, Ристо Лазаров, Ива Петреска	Community 3
Адем Гајтани, Веле Смилевски	Community 4
Матеја Матевски	Community 0 (new, unrelated to the old one)

Cost: the new partition is less seed-stable. Repeating the 30-seed sweep from §2 on the restricted graph gives a markedly lower stability floor than the full graph had:

Graph	Min ARI	Mean ARI	Median ARI
Full graph (incl. `author_similarity`), §2	0.515	0.877	0.906
Restricted graph (word-mediated only)	0.300	0.581	0.541

This is the expected price of the fix, not a sign it is wrong: the author-similarity edges were direct, high-weight author↔author connections that gave Louvain unambiguous structure to lock onto: precisely the "glue" that made Community 0 so artificially stable (§2.5). Without it, the remaining word-mediated graph is sparser, and Louvain's well-known sensitivity to node-visit order shows up more. Per-community confidence is uneven: averaging pairwise co-clustering across the 30 seeds by the seed $= 42$ community gives 0.88 for Community 4 and 0.74 for Community 3 (high confidence) down to 0.49 for Community 2 and 0.27 for the 3-author Community 0 (Матевски, Линдро, Цветкоски — low confidence, on par with a coin flip for some pairs).

Recommendation. Treat the 5-community, word-mediated partition as the current best estimate of stylistic community structure, but do not read the exact random_state $= 42$ assignment as ground truth for the weaker communities (2 and the new 0) the way one might have been tempted to for the old, artifactually-stable Community 0. A seed-ensemble/consensus partition would give a more defensible per-pair confidence than any single seed and is a natural next step if community membership starts being used for a downstream decision (e.g. cluster-based candidate pooling) rather than reported descriptively.

Addendum (poem-id migration). The table above and its seed=42 assignments were computed before a later fix gave every poem a stable `poem_id`, correcting ∼15 `(author, song_title)` pairs in the corpus that had been sharing a title across distinct physical poems (see `CLAUDE.md`'s Data quality context). That fix nudges `pos_transitions.json` and the SKG's edge weights by a small, legitimate amount for the affected authors — among them Блаже Конески, Ефтим Клетников, Влада Урошевиќ, and Пијан Славеј. Re-running `graph_analytics.py` after the fix shows this small a perturbation is enough to move Конески out of his former group with Јаневски/Шопов/Рацин and into the group with Гајтани/Смилевски instead, while Урошевиќ/Славеј/ Клетников and Андреевски/Ѓузел/Лазаров/Петреска remain grouped together (under new community numbers — Louvain's own labels are not stable across runs and carry no meaning by themselves). This is not a contradiction of this section's finding; it is a further, unplanned demonstration of it: this restricted graph's community structure is sensitive enough that even a small, targeted correctness fix elsewhere in the pipeline reshuffles the weaker communities. Properly updating the table above would mean repeating the full 30-seed sweep on the corrected graph, not just swapping in one new run's numbers — that re-verification is out of scope for the poem-id migration and is a natural next step before this section's exact figures are relied on again.

3 Correctness fixes found during validation

Two bugs, unrelated to the tagger switch, were found and fixed while validating pipeline outputs.

3.1 Rhyme-ending extraction

`style_profiler.py`'s `top_rhyme_endings` field took the literal last three characters of each line's last whitespace-split token, with no punctuation stripping. Macedonian poetry in this corpus frequently ends

lines on a standalone em/en dash (a caesura marker) or on punctuation glued to the last word, so every one of the 24 authors' top-5 rhyme endings were contaminated with punctuation artifacts — e.g. Блаже Конески's were `--|на,|на.|ам.|ат.` and Кочо Рацин's were `--|---|ата|ите|ни,`. This was corpus-wide, not an isolated case.

The fix extracts a genuine line-final syllable: it walks backward past trailing pure-punctuation tokens, strips edge punctuation from the real last word, and returns the substring from the onset of the final vowel's syllable (the character after the second-to-last vowel) through the end of the word. After the fix, no author's top rhyme endings contain punctuation, and the endings reflect real Macedonian grammatical suffixes (`-та`, `-те`, `-на`, `-ва`, and similar).

3.2 Style-transfer legacy-fallback tier

`candidate_selection.py`'s last-resort candidate tier (`tier_legacy_fallback`) imported a hardcoded `TARGET_AUTHOR` constant from `llm_probe.py` instead of using the `target_author` parameter it was already given, so whenever this tier fired for any author other than that hardcoded default, it silently supplied *that other author's* TF-IDF vocabulary as replacement candidates. The fix threads `target_author` through to `load_target_words()`, which now accepts an explicit author rather than reading the module constant.

To measure real-world impact, `style_transfer.transfer_style()` was re-run over 960 sampled (source poem, target author) pairs spanning all 24 authors, instrumented to record which tier produced each of the 20,940 replacement slots evaluated:

Tier of the candidate chosen in the output	Share of slots
`primary`	94.20%
`relaxed_semantic`	5.73%
`legacy_fallback`	0.062% (13 slots)
`cross_author_similar`	0.014%

The bug was real wherever it fired, but it fired rarely — 13 of 20,940 slots, concentrated in a handful of sparse-vocabulary authors (8 of the 13 for a single author, Јосип Коцев). It was corrupting a rare edge case, not a meaningful share of style-transfer output.

4 Metric additions to the author style profiles

4.1 Length-normalized lexical diversity

Plain type-token ratio (TTR) in `author_style_profiles.csv` is not length-normalized, and per-author corpus size varies 37-fold across the 24 authors (232 to 8,538 content tokens). On this corpus, plain TTR correlates with token count at Pearson $r = -0.76$ ($p = 1.7 \times 10^{-5}$) — the expected artifact of TTR mechanically falling as a corpus grows, rather than a signal of authorial lexical richness. A Moving-Average Type-Token Ratio (MATTR; Covington and McFall, 2010) column was added, averaging the TTR of every 200-token sliding window over each author's lemma sequence (window size chosen to fit inside the smallest eligible author's 232-token corpus). MATTR against token count: $r = -0.037$ ($p = 0.86$) — no residual length dependence. The original `type_ token_ratio` column is retained for reference but is now documented as diagnostic-only; `mattr` is the column to use for cross-author comparison.

4.2 Stanza-level structure

`author_style_profiles.csv` previously had no stanza-level features. Poem text in `stripped_songs.csv` uses blank-line-delimited stanzas (the same convention already relied on elsewhere in the pipeline for stanza extraction). Two fields were added, computed per author from that delimiting: `avg_stanzas_per_poem` and `avg_lines_per_stanza`. Values are plausible and vary meaningfully across authors — e.g. Ристо Лазаров averages 7.23 stanzas/poem at 2.95 lines/stanza (short, couplet-like stanzas), while Лилјана Дирјан averages 1.14 stanzas/poem at 15.62 lines/stanza (long, largely unbroken poems).

5 Worked example: style transfer after the fixes

As a concrete illustration of where the pipeline stands after the fixes in this note, this section walks through one full style-transfer run on the current (post-fix) pipeline state: Кочо Рацин's "Балада за непознатиот" ("Ballad of the Unknown [Soldier]"), rewritten in Блаже Конески's style via `style_transfer.transfer_style(text, `Блаже Конески')`. This is a real, reproducible pipeline run, not a hand-picked or edited example — it was independently regenerated against the current corpus and model files and diffs byte-identical (modulo typographic quote-mark style) to the run quoted below.

Натаму – в поле битолско
чемрее врба проклета –
под врбата незнаен гроб,
в гроб лежи војник непознат.

Лежи од војна света,
лежи — и веќе земјосал —
силно го тага гласи
мрзне тука загинал.

Никој крај него немаше –
вишното небе врз него,
земјата скришна под него,
над гробот врба спремам.

(…)

И од ним – самовилите
една по една идеа –
од гроб до гроб го мрзне
јунак до јунак – на оро.

(…)

(Full output omitted here for space; see the corpus entry for the source text and re-run `style_transfer.transfer_s` for the complete transferred poem.)

What this demonstrates, and what it doesn't. This call did not pass `source_author`, so only replacement criterion (b) — a lemma absent from the target's vocabulary for a POS in `B_ELIGIBLE_POS` = {ADJ, ADV, VERB} — is active; criterion (a) (source-distinctive/target-absent, any content POS) needs a source author and did not fire here. This is visible directly in the output: NOUN tokens like `гроб` ('grave') and `гробишта` ('graveyard') are preserved verbatim throughout, exactly as `style_transfer.py`'s own design comment (labelled "Criterion (b)") intends — nouns carry subject matter, not style, and are deliberately left alone under this criterion. Adjectives and verbs are substituted where Конески's own vocabulary lacks the Рацин original: `војна световна` ('worldly/mundane war') becomes `војна света` ('holy war'), and `тага изела` ('grief consumed [him]') becomes `тага гласи` ('grief speaks/sounds'), each drawing a real Конески-attested word of the same POS into the same grammatical slot.

Two honest limitations visible in this same output. First, задека ('because') — a conjunction — is itself replaced (with мрзне, 'freezes'), which looks wrong until you check the tagging: CLASSLA tags this token VERB in this context (Aspect=Perf|Tense=Past), not CONJ/SCONJ, so it passes through CONTENT_POS and becomes a legitimate replacement target under the (mistaken) tag. This is the same category of tagging error pos_flags.csv's upos_xpos_clash/unstable_ tag reasons are built to catch for common cases — задека was not one already in data/pos_corrections.csv at the time of this run, so it slipped through uncorrected. Second, дигаа ('they raised', Number=Plur, 3rd-person imperfect) is replaced by мрзне, which surfaces in an unmarked/singular-present-looking form — the morphological lookup evidently had no attested Конески plural-imperfect form of that lemma to realise into, and fell back to a lower morph tier (nearest_attested or unresolved) rather than the requested agreement. Neither of these breaks the pipeline, and neither is new to this session's fixes — they are exactly the kind of residual, low-frequency noise the tier/morph-tier logging in candidate_selection.py exists to make visible and auditable, rather than silent.

6 Prompt-construction fixes in the LLM style-transfer pipeline

A second, independent style-transfer approach (llm_style_transfer.py) sends a single LLM prompt — an LLM-written summary of the source poem, its content keywords, the target author's style profile in prose, a relevance-filtered vocabulary palette, tone exemplar lines, and structural targets — and asks the model to write a new poem, rather than substituting tokens into masked slots. Auditing the per-run records it writes to data/llm_transfer_logs/ surfaced three defects in how that prompt was assembled. None of them raised an error; all of them degraded the prompt silently.

6.1 Duplicate exemplar lines

The tone-exemplar block repeated lines verbatim. In a Ацо Шопов run, four of nine exemplars were two duplicated pairs (Треба да бидеме подобри. and Вечер е тиха, дремлива,), and a further two came from a single poem. The block is introduced to the model as a rhythm reference, so a verbatim repeat inside it is an instruction to repeat; the generated poem duplicated whole stanzas and repeated single lines three times.

Duplicates entered by two routes. First, exemplar_selection.py's qualifying_lines() capped candidates at two lines per poem but did not check whether they were the same text, so a refrain spent the entire quota on one line — Треба да бидеме подобри opens with that line twice (poem_id 88), and Партизанска колона repeats Вечер е тиха, дремлива, at lines 2 and 21 (poem_id 50). Because identical text yields an identical line embedding, both copies always landed in the same k-means cluster and the two-per-cluster rule selected both. Second, the corpus's near-duplicate poems each contributed the same line independently: Блаже Конески's Бура (poem_id 13) and БУРА (poem_id 566) are one poem under two title casings, and 24 such author/lowercased-title groups exist.

The fix de-duplicates normalized line text (case-folded, whitespace-collapsed) author-wide rather than per-poem, which closes both routes; a repeat is skipped without consuming a per-poem slot, so a refrain poem still contributes two distinct lines. Separately, cluster_and_select() now enforces one line per source poem globally across clusters rather than within them, and processes clusters smallest-first so that a constrained cluster claims its poem before a roomier one takes it. Across the corpus's 38 authors this costs 7 exemplar lines in total (218 to 211) and no author falls below 5. Exemplar records now carry poem_id alongside song_title, since the title alone cannot distinguish Бура from БУРА in a run log.

6.2 Summary language

Summaries were requested in Macedonian and returned ungrammatical: the logged summary for Кочо Рацин's `Балада за непознатиот` opened `Песната приказува` (for `прикажува`), and continued `го тага неговото загинување` and `го поседуваат несигурниот смисол на нивната смрт` — text that conveys the gist but is not well-formed Macedonian. The local model in use (`qwen3:14b`) reads Macedonian considerably better than it writes it, and the summary is scaffolding consumed by the generation step, not output shown to a reader.

Summaries are therefore now taken in English. The prompt's lexical material — content keywords, vocabulary palette, exemplar lines — and the generated poem itself remain Macedonian; only the description of the source poem's content changes language. This matches the existing choice to phrase instructions in English (`DEFAULT_INSTR_LANG`). `SUMMARY_PROMPT_VERSION` was bumped to 2, which invalidates cached v1 summaries via the cache key.

6.3 Unstated output language

The prompt never instructed the model which language to write in. Under v1 this went unnoticed because the Macedonian summary was the largest Cyrillic block in the prompt and the model followed it by example. Moving the summary to English removed that implicit cue, so the instruction is now explicit (*"Write the poem in Macedonian, in Cyrillic script"*). The same audit found that `--lang mk` was an advertised CLI choice with no corresponding entry in `PROMPT_TEMPLATE`, so it raised `KeyError` during prompt assembly; the choices are now derived from the templates that actually exist.

6.4 A lemmatization error surfaced by the prompt

Reading the assembled prompts also caught a tagging error that no flag rule raises. The content keywords for Кочо Рацин's `Балада за непознатиот` included `петлин`, which is not a Macedonian word: classla lemmatizes the definite plural `петлите` to `петлин` rather than `петел` ('rooster'). It gets every other form of the word right — `петел`, `петелот` and `петли` all lemmatize to `петел` — so the corpus carried one lemma split across two spellings, 6 tokens under the wrong one, across three authors (Кочо Рацин, Милош Линдро, Ефтим Клетников).

Nothing in `pos_flags.csv` catches this: the rules look at tags, not at lemma consistency, and the POS and `xpos` (`Ncmpny`) were both correct here. The fix is a lemma-only entry in `data/pos_corrections.csv` — blank `correction`, `lemma_ fix` of `петел`, corpus-wide scope — which `corrections.apply_to` applies without clearing `xpos`/`feats`, since no POS judgement was overruled.

The effect on the prompt is not simply a substituted keyword. Merging the 6 tokens into `петел`'s existing count raises that lemma's corpus frequency, lowering its IDF, and it consequently drops out of the poem's top-10 keywords altogether — the previous ranking was inflated by a spurious singleton lemma. Lemma-keyed artifacts further downstream (`word_embeddings.json`, the SKG, `morph_lookup.json`) still carry a stale `петлин` entry; at 6 tokens in 70,531 it was not judged worth a full rebuild, and `петел` already has an embedding, so the keyword remains embeddable.

6.5 Switching the generation model

All of the prompt work above was developed and logged against `qwen3:14b` running locally under Ollama. With the prompt defects fixed, that model was the remaining constraint on output quality. It writes Macedonian poorly: generated poems repeated whole stanzas, ran single lines three times (`и се вика: ``Иде, иде, иде, иде, иде.''`), and produced non-words and broken agreement (`кое се вик`, `каде се вод`).

The pipeline was therefore switched to `gemma-4-31b-it`, a substantially larger model with much stronger Macedonian. It is now the default (`DEFAULT_MODEL`/`DEFAULT_BACKEND` in `llm_style_transfer.py`); Ollama and `qwen3:14b` remain reachable through `--backend ollama --model qwen3:14b`, so the earlier logs stay reproducible.

Two practical obstacles shaped how it is reached. First, every OpenRouter `:free` Gemma route answers HTTP 429 from a shared upstream pool at Google AI Studio, so the model is called through Google's Generative Language API directly, with its own key — a third `google` backend in `llm_client.py` alongside `ollama` and `openrouter`. Second, Gemma 4 always reasons before answering, and cannot be stopped from doing so: `generationConfig.thinkingConfig.thinkingBudget` is rejected with `400 Thinking budget is not supported for this model`. The scratchpad returns as a response part flagged `thought=true` beside the answer part, so it has to be filtered out — concatenating every part yields the model arguing with itself rather than the poem.

Thinking tokens are charged against `maxOutputTokens`, and their number varies enough between runs of the *same* prompt that no fixed budget is safe: observed figures for one style-transfer prompt were 3231 thought tokens (305 for the poem), then 15,997 with no answer produced at all, then 2075 on a re-send. The backend therefore requests the model's full 32,768-token output ceiling and retries once when a response comes back with thinking but no answer part, rather than attempting to budget around the variance.

6.6 Effect

Re-running both logged transfers from the same source poem confirmed the prompt-level fixes: exemplar blocks now contain no repeated line and no repeated source poem (8 lines/8 poems for Ацо Шопов, 10/10 for Блаже Конески), and the summary is fluent, accurate English.

The model switch is visible in the structural fit, which was measured but never targeted by either change:

Structural target (same prompt, same source poem)	`qwen3:14b`	`gemma-4-31b-it`
Ацо Шопов, lines (target 20.7)	16	22
Ацо Шопов, Δ tokens per line	$+0.99$	$+0.10$
Блаже Конески, lines (target 16.4)	14	16
Блаже Конески, stanzas (target 2.3)	2	2

The qualitative gap is larger than the table suggests. Gemma places the target author's palette words in grammatical context rather than stacking them — `тишината личи на тешка небиднина` uses Шопов's own coinage `небиднина` in a well-formed clause — and for Конески it produced a strict $8 + 8$ poem built on a consistent adjective–adverb–verb line template (`студена коска тесно го стега`, `чемреена мака бавно го јаде`), which is the adjective-before-noun pattern that author's style profile describes.

Content preservation improved as well, and traceably so: the English summary identifies that the dead soldier *refuses* to join the heroes' oro because he lacks their honour. That refusal is the source poem's central turn, it is absent from the Macedonian v1 summary, and it appears in both regenerated poems.

7 Distillation and reinforcement learning

The downstream work that builds on the pipeline above — distilling the Gemma style-transfer prompt into a LoRA-tuned Qwen3-4B student, its evaluation, the GRPO stage (reward design, v1 and v2 results) and the Gemma 4 E4B second student — is reported separately in `rl_report.tex`.

8 Reproducibility

Figures in §2–§2.5 (the diagnosis) were produced against `skg_final.gexf` and `graph_analytics.csv` as of commit `41bb49e` (or later, up to the fix below, on branch `feats-morph-lookup`), and against the pre-switch versions recoverable at commit `5f03983`/`363009a`. Figures in §2.6 (the fix) were produced against `src/graph_analytics.py`'s current working-tree state on that same branch, which restricts Louvain's input graph to word-mediated edges; commit this change alongside the CSV it produces to keep the two in sync. The Louvain seed-stability sweeps and the tier-frequency measurements were produced by one-off scripts run against the live pipeline rather than checked-in tests, since they characterize algorithmic stability and empirical tier frequency rather than fixed expected behavior.